TECHNICAL SPECIFICATION & ENGINEERING MANUAL

SmartAttendance™ Hardware & Firmware Reference

Low-Level Subsystem Architecture, Pinout Matrix, MIPI CSI/DSI Drivers, ESP-DL Neural Inference, Power Topology, FreeRTOS Concurrency & Board Diagnostics

- **SoC & Memory:** Espressif ESP32-P4 RISC-V Dual-Core @ 400 MHz + 32MB Hex-PSRAM @ 200 MHz
- **Physical Controls:** Left-side slide switch (UP=ON, DOWN=OFF), USB-C charging slot (5V 2-3A)
- **Display Bridge:** EK79007 MIPI-DSI (2 lanes @ 900 Mbps, 51 MHz DPI clock, 1024×600 IPS panel)
- **Touch Controller:** GT911 Capacitive Touch with verified manual GPIO40/42 I2C address lock (0x5D)
- **Camera Subsystem:** SC2336 24-Pin MIPI-CSI (SCCB 0x36) at top of screen, hardware ISP Bayer BGGR
- **DMA & Concurrency:** Double-buffered frame acquisition in PSRAM (`s_detection_fb_buf`) eliminating tearing
- **Edge AI Models:** Quantized int8 ESP-DL FaceDetect (MSR/MNP) & MobileFaceNet 128D embedding extractor
- **Power & Gas Gauge:** Dual power rails (LDO3 2.5V, LDO4 3.3V), STC8 battery fuel gauge (no audio/speaker)
- **Storage Architecture:** SDMMC 1-bit FAT32 on bottom slot (`/sdcard`) with SQLite3 DB (DO NOT REMOVE!)

Target Audience:	Embedded Engineers, Firmware Devs, Hardware Techs	**Hardware Revision:**	CrowPanel Advanced 7″ ESP32-P4 v1.0
Framework / SDK:	ESP-IDF v5.3+ / ESP-DL v1.1.4+	**Typography:**	Times New Roman (12pt Body)
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[!] HARDWARE WARNING: Confidential Engineering Document

This manual contains low-level hardware schematics, register configurations, and flash partition offsets intended exclusively for technical personnel. End-user operating guidelines are documented in the companion SmartAttendance User Manual.

Engineering Table of Contents

Section 1: Hardware Architecture & Electrical Pinout	Detailed pin mapping, power rail sequencing (LDO3/4), bus interfaces, slide switch
Section 2: MIPI-DSI Display & Backlight Subsystems	EK79007 bridge parameters, LVGL 9 SPIRAM buffers, GPIO31 PWM
Section 3: GT911 Capacitive Touch Controller Quirks	Manual GPIO reset sequence, 16-bit register commands, wake-up sequence
Section 4: MIPI-CSI Camera Driver & ISP Pipeline	SC2336 24-pin CSI at top of screen, SCCB 0x36, Bayer BGGR, exposure timing, DMA buffer
Section 5: High-Speed PSRAM & Flash Memory Architecture	Hex-mode PSRAM @ 200MHz, experimental symbols, heap allocation caps
Section 6: Power Supply, Battery Telemetry & STC8 Gauge	Left-side USB-C 5V 2-3A, ADC voltage sense, STC8 UART packets, color states (no audio)
Section 7: Deep Learning Vision Engine & Face Embeddings	ESP-DL inference, 5-point landmark warping, MobileFaceNet 128D int8 vectors
Section 8: Multi-Frame Temporal Consensus & Margin Gating	6-frame window, 3-frame early exit, dynamic margin delta, anti-glare math
Section 9: FreeRTOS Task Architecture & Mutex Locking	Task priorities, stack sizing, UI recursive mutex safety, deadlock avoidance
Section 10: Partition Table & Dual-Bank OTA Flashing	6MB app partitions, factory fallback, esp_ota routines, flashing scripts
Section 11: SQLite3 Schema & On-Disk Data Structures	Bottom MicroSD card (`attendance.db`), index optimization, backup mirroring (DO NOT REMOVE!)
Section 12: Board-Level Diagnostics & Hardware Repair	Multimeter test points, FPC connector troubleshooting, recovery flashing

Section 1: Hardware Architecture & Electrical Pinout

The Elecrow CrowPanel Advanced 7″ ESP32-P4 v1.0 integrates the ESP32-P4 SoC with external Hex-PSRAM, MIPI CSI/DSI bridges, and power management circuits. Physical layout verified from hands-on testing:

- **Power Switch:** Physical slide switch located on the **left edge** (slide UP to connect battery/power, slide DOWN to disconnect).
- **Power / Charging Input:** USB Type-C port located on the **left edge directly below the slide switch**. Recommended power supply: **5.0V, 2.0A to 3.0A DC**.
- **Camera Module:** OmniVision SC2336 mounted along the **top edge of the screen** via 24-pin ribbon cable.
- **MicroSD Storage:** High-speed SDMMC slot located on the **bottom edge of the board**. **MANDATORY: DO NOT REMOVE DURING OPERATION!** The slot holds attendance.db and AI model weights.
- **Audio Subsystem Notice:** The audio amplifier and speaker have been permanently stripped from firmware v1.2 (2026-06-11). Audio defines in config.h are inactive tombstones.

Hardware Pinout & Bus Mapping Matrix

Peripheral	GPIO Pin	Interface / Net Name	Engineering Notes
Backlight PWM	GPIO 31	LEDC PWM Channel 0	30 kHz carrier frequency, 11-bit resolution (duty 200-2047).
Touch SDA	GPIO 45	I2C_NUM_0 SDA	Requires external 4.7kΩ pull-up to 3.3V.
Touch SCL	GPIO 46	I2C_NUM_0 SCL	100 kHz standard clock speed with pull-up.
Touch Reset	GPIO 40	GT911_RST	Must be driven via manual GPIO pulse to fix 0x5D address.
Touch Interrupt	GPIO 42	GT911_INT	Hold low during RST release to latch address 0x5D.
Camera CSI Data	DSI/CSI PHY	MIPI-CSI 2-Lane	OmniVision SC2336 24-pin FPC ribbon; RAW8 format at top of screen.
Camera SCCB SDA	GPIO 7	SCCB I2C SDA	Sensor address 0x36 (alternate address; NOT 0x30).
Camera SCCB SCL	GPIO 8	SCCB I2C SCL	Clock for camera sensor register read/write.
SDMMC Clock	GPIO 43	SD_CLK	SDMMC 1-bit clock line; 20 MHz to 40 MHz clock (bottom slot).
SDMMC Command	GPIO 44	SD_CMD	SDMMC command line with internal pull-up enabled.
SDMMC Data 0	GPIO 39	SD_D0	SDMMC data 0 bidirectional transfer.
Battery ADC	GPIO 4	ADC_CHANNEL_4	Monitors 3.3V to 4.2V battery via resistor divider network.
DS1302 RTC CLK	GPIO 51	J7 Header IO51	Serial clock for external battery-backed RTC.
DS1302 RTC DAT	GPIO 50	J7 Header IO50	Bidirectional serial data line for RTC register transfers.
DS1302 RTC CE	GPIO 49	J7 Header IO49	Chip Enable (active high) for DS1302 bus access.

[X] CRITICAL CONSTRAINT: Power Rail Sequencing (LDO3 & LDO4)

The EK79007 display bridge and MIPI PHY require internal LDO3 (2.5 V) and LDO4 (3.3 V) rails. If firmware fails to initialize these PMU regulators prior to display driver initialization, the MIPI-DSI link will fail to negotiate clock synchronization, leaving the panel permanently dark.

Section 2: MIPI-DSI Display & Backlight Subsystems

The display panel connects via a 2-lane MIPI-DSI interface managed by an EK79007 driver IC. Key physical parameters:

- **Lane Bit Rate:** 900 Mbps across 2 differential data lanes.
- **DPI Pixel Clock:** 51 MHz nominal.
- **Resolution & Porch Timing:** 1024×600 active pixels. HSYNC pulse: 20, HBP: 140, HFP: 160. VSYNC pulse: 3, VBP: 20, VFP: 12.
- **Critical Driver Constraint:** Do **NOT** call esp_lcd_panel_disp_on_off(). The EK79007 bridge does not support software display power toggle; attempting to do so locks the MIPI PHY.
- **LVGL 9 Optimization:** Allocate draw buffers strictly from PSRAM: swap_bytes=false, buff_spiram=true, sw_rotate=true, double_buffer=true.
- **Backlight Formula:** LEDC 30 kHz on GPIO31. Duty formula: duty = (pct != 0) ? ((pct * 18) + 200) : 0.

Section 3: GT911 Capacitive Touch Controller Quirks

The Goodix GT911 capacitive touch IC requires a precise hardware reset timing sequence to latch its I2C slave address at 0x5D. Standard generic drivers fail because they do not control RST and INT with sub-millisecond precision.

```
// 1. Install legacy I2C driver on I2C_NUM_0 (GPIO 45 SDA, GPIO 46 SCL, 100 kHz)
i2c_config_t i2c_conf = {
    .mode = I2C_MODE_MASTER, .sda_io_num = 45, .scl_io_num = 46,
    .sda_pullup_en = GPIO_PULLUP_ENABLE, .scl_pullup_en = GPIO_PULLUP_ENABLE,
    .master.clk_speed = 100000,
};
i2c_param_config(I2C_NUM_0, &i2c;_conf);
i2c_driver_install(I2C_NUM_0, I2C_MODE_MASTER, 0, 0, 0);

// 2. Hardware timing pulse to force address 0x5D:
gpio_set_direction(40, GPIO_MODE_OUTPUT); // RST
gpio_set_direction(42, GPIO_MODE_OUTPUT); // INT
gpio_set_level(40, 0); gpio_set_level(42, 0);  // Both driven low
vTaskDelay(pdMS_TO_TICKS(1));
gpio_set_level(40, 1);                          // Release RST high while INT stays low
vTaskDelay(pdMS_TO_TICKS(50));
gpio_set_level(42, 1);                          // Release INT high -> Latches 0x5D address!
vTaskDelay(pdMS_TO_TICKS(20));

// 3. Configure esp_lcd_panel_io_i2c (16-bit register commands are MANDATORY):
esp_lcd_panel_io_i2c_config_t io_cfg = {
    .dev_addr = 0x5D, .control_phase_bytes = 1, .dc_bit_offset = 0,
    .lcd_cmd_bits = 16,     // MUST be 16, NOT 8!
    .lcd_param_bits = 16,   // MUST be 16, NOT 8!
    .flags.disable_control_phase = 1,
};

// 4. Touch driver configuration (RST and INT set to NC because handled above):
esp_lcd_touch_config_t tp_cfg = {
    .x_max = 1024, .y_max = 600,
    .rst_gpio_num = GPIO_NUM_NC, .int_gpio_num = GPIO_NUM_NC,
    .flags = { .swap_xy = 0, .mirror_x = 1, .mirror_y = 1 },
};
esp_lcd_touch_new_i2c_gt911(tp_io, &tp;_cfg, &tp;);
vTaskDelay(pdMS_TO_TICKS(1000));
esp_lcd_touch_exit_sleep(tp); // Mandatory GT911 wake-up command
```

Section 4: MIPI-CSI Camera Driver & ISP Pipeline

The camera sensor is an OmniVision SC2336 connected to the ESP32-P4 hardware ISP via 2-lane MIPI-CSI at the top of the display:

- **Sensor Mount:** Positioned along the top display bezel, pointing forward toward passing users.
- **SCCB Bus Address:** 0x36 (7-bit address; not 0x30).
- **Bayer Pixel Format:** Native sensor output is **BGGR** RAW8. Do NOT enable HMIRROR without switching ISP Bayer order to GBRG, or severe chromatic distortion will result.
- **Resolution & Framerate:** MIPI_2lane_24Minput_RAW8_1280x720_30fps (the only hardware-supported profile on this board revision).
- **Indoor Exposure Baseline:** exposure_us = 15000, exposure_val = 1100, gain = 4000.
- **Double-Buffering DMA Protection:** A dedicated buffer s_detection_fb_buf is allocated in PSRAM during boot. When the detection task is ready, the camera task copies the frame atomically, ensuring the face detector and landmark alignment never read a partially overwritten buffer.
- **Camera Profiles:** Firmware provides 5 runtime exposure presets selectable via UI buttons: Auto, Outdoor, Indoor, Bright, and Low Light.

Section 5: High-Speed PSRAM & Flash Memory Architecture

The ESP32-P4 requires 32MB Hex-mode PSRAM running at 200 MHz to support simultaneous 720p MIPI video decoding, LVGL frame rendering, and neural network tensor allocations.

```
# Mandatory symbols in sdkconfig / sdkconfig.defaults.esp32p4:
CONFIG_IDF_EXPERIMENTAL_FEATURES=y          # Required to unlock 200MHz PSRAM
CONFIG_SPIRAM=y
CONFIG_SPIRAM_MODE_HEX=y                    # 16-line Hex-mode bus
CONFIG_SPIRAM_SPEED_200M=y                  # 200 MHz clock rate
CONFIG_SPIRAM_USE_MALLOC=y
CONFIG_SPIRAM_ALLOW_BSS_SEG_EXTERNAL_MEMORY=y
CONFIG_SPIRAM_RODATA=y
CONFIG_ESPTOOLPY_FLASHSIZE_16MB=y           # 16 MB QSPI Flash
CONFIG_I2C_SKIP_LEGACY_CONFLICT_CHECK=y    # Avoids driver/i2c legacy symbol clash
```

[X] CRITICAL CONSTRAINT: Stack Allocation Anti-Pattern

Never allocate large frame arrays on the FreeRTOS task stack (e.g., uint8_t crop[112*112*3]). Doing so causes an immediate task stack overflow. All vision tensors and image crops must be dynamically allocated from PSRAM using heap_caps_malloc(size, MALLOC_CAP_SPIRAM) and freed upon task completion.

Section 6: Power Supply, Battery Telemetry & STC8 Gauge

The power management system combines continuous USB-C DC input with single-cell LiPo battery backup:

- **Mechanical Power Switch:** Slide switch on left edge (UP connects battery/DC, DOWN disconnects power).
- **DC Input & Charging:** USB Type-C port on left edge below switch. Requires **5V, 2A to 3A** for proper simultaneous operation and cell charging.
- **ADC Sensing Network:** GPIO 4 (ADC1 Channel 4) measures battery terminal voltage through a 2:1 resistive divider (3.30V empty to 4.20V full).
- **STC8 Gas Gauge Protocol:** Onboard STC8 coprocessor samples voltage and communicates percentage to the ESP32-P4.
- **UI Color Display Implementation:**
  - **Charging:** Displays LV_SYMBOL_CHARGE and percentage in **Cyan Blue** (#00CFFF).
  - **Normal (≥15%):** Displays icon and percentage in **Theme White/Black** (NO green color, NO bar meter).
  - **Low Battery (<15%):** Icon and percentage turn solid **Red** (#FF4444).
  - **5% Emergency Shutdown:** Calls db_sync_to_disk(), unmounts /sdcard, and safely halts to protect database integrity.
- **Calibration Logger:** When `BATTERY_CALIB_LOGGING=1` in `config.h`, records ADC mV and STC8 percentage to `/sdcard/logs/batt_calib.csv` every 20 seconds.

Section 7: Deep Learning Vision Engine & Face Embeddings

The biometric engine uses Espressif ESP-DL quantized neural network models executed on the RISC-V SIMD architecture:

- **Model Locations on SD Card:**
  - /sdcard/models/p4/human_face_detect_msr_s8_v1.espdl (Fast multi-scale face proposal)
  - /sdcard/models/p4/human_face_detect_mnp_s8_v1.espdl (5-point keypoint landmark detection)
  - /sdcard/models/p4/mfn_s8_v1.espdl (MobileFaceNet 128D int8 embedding extractor)
- **Keypoint Indexing:** ESP-DL outputs 5 landmarks in order: [Left Eye, Left Mouth, Nose, Right Eye, Right Mouth]. Landmarks are copied 1-to-1 without reordering to keep yaw angles accurate.
- **Affine Transform:** Warps cropped face keypoints to fixed coordinate anchors in a canonical 112×112 RGB565 matrix.
- **Feature Extraction:** MobileFaceNet outputs a 128-dimensional signed int8 vector e □ [-128, 127]^128.

Section 8: Multi-Frame Temporal Consensus & Margin Gating

To achieve near-zero false acceptance rates without sacrificing throughput:

- **Sliding Window Accumulator:** Tracks up to 6 valid frames over a 1500ms window.
- **Early Exit Fast Path:** If 3 consecutive frames show unanimous candidate matching with similarity S ≥ 0.80, verification completes immediately.

- **Dynamic Margin Gate:** Required separation margin over runner-up candidate #2 depends on Top-1 similarity:
  - **S ≥ 0.78:** Requires ΔS ≥ 0.03.
  - **0.74 ≤ S < 0.78:** Requires ΔS ≥ 0.05.
  - **0.70 ≤ S < 0.74:** Requires strict ΔS ≥ 0.08.
- **Visual-Only Feedback:** Results are displayed on screen via ui_show_attendance_feedback() (green banner `#003D1A` for success, orange `#3D2000` for duplicates, red text for unknown). No audio chime is emitted.
- **Quality Gates:** Specular glare filter (rejects frames with >15% pixels Y>240) and mouth elongation gate (rejects frames with ratio > 0.85).

Section 9: FreeRTOS Task Architecture & Mutex Locking

The firmware executes across two symmetric RISC-V cores. Strict task priority and synchronization rules prevent race conditions:

Task Name	Priority	Stack Size	Core Affinity & Architectural Role
LVGL UI Task	10 (Highest)	8,192 bytes	Core 1. Handles screen rendering and touch event dispatch.
Detection Task	8	32,768 bytes	Core 0. Runs ESP-DL neural face detector and landmark alignment.
Camera Task	7	8,192 bytes	Core 0. Receives MIPI DMA interrupts and manages frame buffers.
Database Task	6	8,192 bytes	Core 1. Executes SQLite3 attendance writes and user queries.
Network Task	5	16,384 bytes	Core 0. Manages Wi-Fi, HTTPS sync API, and Telegram bot calls.
Battery Task	4	4,096 bytes	Core 1. Periodic 10s ADC voltage and STC8 telemetry polling.

LVGL Thread Safety Rule: Any non-display task invoking LVGL widget functions must acquire ui_acquire(). If ui_acquire() returns false (timeout), the caller MUST skip the UI call and bypass ui_release() to avoid corrupting the FreeRTOS recursive semaphore.

Section 10: Partition Table & Dual-Bank OTA Flashing

The 16 MB flash memory is partitioned for safe dual-bank A/B firmware upgrades:

```
# Name,     Type, SubType, Offset,   Size,     Flags
nvs,        data, nvs,     0x9000,   0x6000,
otadata,    data, ota,     0xf000,   0x2000,
phy_init,   data, phy,     0x11000,  0x1000,
ota_0,      app,  ota_0,   0x20000,  0x600000, # 6 MB Application Bank 0
ota_1,      app,  ota_1,   0x620000, 0x600000, # 6 MB Application Bank 1
storage,    data, spiffs,  0xc20000, 0x3d0000,
```

Flashing Commands:

• **Full Clean & Build:** idf.py fullclean && idf.py build

• **Flashing via COM port:** esptool.py --chip esp32p4 -p COM5 -b 460800 --before default_reset --after hard_reset write_flash 0x0 build/bootloader/bootloader.bin 0x10000 build/partition_table/partition-table.bin 0x20000 build/SmartAttendance.bin

Section 11: SQLite3 Schema & On-Disk Data Structures

Biometric profiles and attendance events are stored in /sdcard/attendance.db on the bottom MicroSD slot. **MANDATORY: DO NOT REMOVE THE CARD!**

```
-- Users Table (Biometric Templates & Biographical Data)
CREATE TABLE IF NOT EXISTS users (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    uuid TEXT UNIQUE NOT NULL,
    name TEXT NOT NULL,
    student_id TEXT UNIQUE,
    phone_number TEXT,
    telegram_id TEXT,
    role TEXT NOT NULL CHECK(role IN ('student','lecturer','admin')),
    face_embedding BLOB,               -- 128-byte int8 MobileFaceNet vector
    created_at INTEGER NOT NULL,
    updated_at INTEGER,
    enroll_quality INTEGER DEFAULT 0,  -- Quality score (0 - 100)
    enroll_accepted INTEGER DEFAULT 0,
    enroll_rejected INTEGER DEFAULT 0,
    model_version INTEGER DEFAULT 1,
    embedding_dim INTEGER DEFAULT 128
);

-- Attendance Log Table
CREATE TABLE IF NOT EXISTS attendance (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER NOT NULL,
    session_id INTEGER,
    course_code TEXT,
    timestamp INTEGER NOT NULL,
    confidence REAL NOT NULL,
    synced INTEGER DEFAULT 0,          -- 0=local only, 1=synced to cloud
    FOREIGN KEY (user_id) REFERENCES users(id)
);
```

Section 12: Board-Level Diagnostics & Hardware Repair

Technicians diagnosing hardware failures should follow this empirical troubleshooting matrix:

Hardware Symptom	Diagnostic Test Point	Root Cause & Repair Procedure
Board fails to enumerate on USB	Check left-side slide switch (must be UP); measure 5V rail at USB-C input.	Slide switch up to connect power. Ensure USB-C power supply provides at least 5V 2A. Check for shorts on 3.3V rail.
Display blank; backlight off	Probe GPIO31 with oscilloscope; check LDO3 (2.5V) and LDO4 (3.3V).	Verify 30 kHz PWM carrier on GPIO31. If PWM is present but backlight is off, check FPC backlight driver boost circuit.
Touch I2C arbitration failure	Probe GPIO45 (SDA) and GPIO46 (SCL) with logic analyzer.	Verify external 4.7kΩ pull-up resistors to 3.3V. Inspect GPIO40 (RST) and GPIO42 (INT) for broken solder joints on 6-pin touch FPC.
Camera probe returns SCCB error	Verify 24-pin FPC latch connection at top edge of screen; probe 3.3V analog rail.	Ensure gold contacts face towards PCB. Confirm sensor address is 0x36. Replace SC2336 camera module if SCCB fails to respond.
MicroSD read timeout / missing DB	Check bottom MicroSD slot; verify GPIO43 (CLK), GPIO44 (CMD), GPIO39 (D0).	Ensure card is firmly inserted in bottom slot (DO NOT REMOVE during operation). Confirm FAT32 format with models/p4/ present.

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